

Energy-Flow Cosmology: Empirical Validation of the EFC Screening Model Against the Radial Acceleration Relation

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Abstract. We present a comprehensive empirical test of the Energy-Flow Cosmology (EFC) screening model against the radial acceleration relation (RAR) using 174 galaxies (3264 data points) from the SPARC database. An unbinned maximum-likelihood fit with intrinsic scatter yields a screening exponent $k = 0.415 \pm 0.029$, a transition acceleration $g^\dagger = (2.51 \pm 0.60) \times 10^{-10} \text{ m/s}^2$, and intrinsic scatter $\sigma_{\text{int}} \approx 0.14$ dex, consistent with RAR literature values. Bootstrap resampling (500 iterations, galaxy-level) preserves intra-galaxy correlations and provides robust uncertainty estimates. The derived phase-gain parameter $C = k/a_G = 4.4 \pm 0.6$ connects galactic and cosmological scales through a single universal coupling $a_G = 0.094 \pm 0.01$, independently determined from the H_0 tension. We extend the analysis to weak gravitational lensing (KiDS-1000 cosmic shear), cluster merger geometry (Bullet Cluster), and the Hubble tension, demonstrating that the EFC framework provides a consistent, multi-scale description from galactic rotation curves to cosmological observables. We present explicit falsification criteria and discuss the transition from “tension resolution” to “precision diagnostics” in light of Stage-III survey convergence.

Keywords: Energy-Flow Cosmology, radial acceleration relation, galaxy rotation curves, SPARC, screening model, weak lensing, KiDS-1000, Bullet Cluster, Hubble tension, modified gravity

Epistemic status: Empirical validation against published data. Fixed-parameter consistency tests. Conjectured closure relations stated with explicit fail conditions.

1 Introduction

The radial acceleration relation (RAR) is one of the tightest empirical correlations in extragalactic astronomy. First systematically characterized by McGaugh, Lelli & Schombert (2016) [1], the RAR connects the observed centripetal acceleration g_{obs} in disk galaxies to the baryonic acceleration g_{bar} predicted from the visible mass distribution alone. The relation holds across four decades in acceleration with remarkably low scatter (~ 0.13 dex), posing a challenge for models that invoke stochastic dark matter halo properties.

Energy-Flow Cosmology (EFC) offers a thermodynamic interpretation: gravitational dynamics are modulated by entropy gradients in the local energy flow. In regimes where the baryonic acceleration falls below a characteristic transition scale $g^\dagger$, the entropy-driven screening produces an enhancement of the effective gravitational coupling. The EFC screening form for the RAR is:

$$\ln \mu = k \ln \left(1 + \frac{g^\dagger}{g_{\text{bar}}} \right), \quad (1)$$

where $\mu = g_{\text{obs}}/g_{\text{bar}}$ is the gravitational enhancement factor, k is the screening exponent, and $g^\dagger$ is the transition acceleration scale. Equivalently, $\mu = (1 + g^\dagger/g_{\text{bar}})^k$.

This paper reports the results of fitting Eq. (1) to the SPARC database and extends the analysis to three additional observational domains: weak gravitational lensing (Section 4), cluster merger geometry (Section 5), and the Hubble tension (Section 6).

2 Data and Method

2.1 SPARC Database

We use the Spitzer Photometry and Accurate Rotation Curves (SPARC) database [2], containing 175 disk galaxies with 3.6 μm photometry and high-quality HI/H α rotation curves. After quality cuts ($Q \leq 2$, inclination $> 30^\circ$), 174 galaxies with 3264 individual data points remain.

For each radial point i , the baryonic acceleration $g_{\text{bar},i}$ is computed from the mass model (gas + disk + bulge components) and $g_{\text{obs},i}$ from the observed rotation velocity via $g_{\text{obs}} = V_{\text{obs}}^2/r$. We define:

$$y_i = \ln \mu_i = \ln \left(\frac{g_{\text{obs},i}}{g_{\text{bar},i}} \right). \quad (2)$$

2.2 Likelihood

We employ unbinned maximum likelihood with intrinsic scatter σ_{int} :

$$\ln \mathcal{L} = -\frac{1}{2} \sum_i \left[\frac{\left(y_i - k \ln(1 + g^\dagger/g_{\text{bar},i}) \right)^2}{\sigma_{y,i}^2 + \sigma_{\text{int}}^2} + \ln(\sigma_{y,i}^2 + \sigma_{\text{int}}^2) \right], \quad (3)$$

where $\sigma_{y,i}$ is the measurement uncertainty on $\ln \mu_i$, propagated from the SPARC-provided uncertainties on V_{obs} and the mass model via $\sigma_y \approx 2\sigma_V/V$.

2.3 Bootstrap Resampling

Uncertainty estimates are obtained via bootstrap resampling (500 iterations). We resample galaxies with replacement and include all radial points per selected galaxy, thereby preserving intra-galaxy correlations. This approach is more conservative than point-level resampling and guards against underestimating parameter uncertainties.

3 Results

3.1 Canonical Fit

Table 1: Summary of fitted and derived parameters.

Parameter	Value	Source
k (screening exponent)	0.415 ± 0.029	SPARC fit (bootstrap)
$g^\dagger$ (transition scale)	$(2.51 \pm 0.60) \times 10^{-10} \text{ m/s}^2$	SPARC fit
σ_{int} (intrinsic scatter)	$0.33 \text{ in } \ln \mu$ ($\approx 0.14 \text{ dex}$)	SPARC fit
a_G (universal coupling)	0.094 ± 0.01	H_0 analysis (indep.)
$C = k/a_G$ (phase-gain)	4.4 ± 0.6	Derived

The intrinsic scatter $\sigma_{\text{int}} = 0.33$ in $\ln \mu$ corresponds to $0.33/\ln 10 \approx 0.14$ dex, consistent with the RAR literature value of ~ 0.13 dex. The screening exponent $k = 0.415$ is robustly constrained within the range $[0.38, 0.45]$ across alternative analysis choices (binned fits, different quality cuts, per-galaxy medians, χ^2 minimization).

3.2 Cross-Scale Consistency

The phase-gain parameter $C = k/a_G$ connects the galactic screening exponent k (from SPARC) to the universal coupling a_G (from cosmological H_0 tension analysis). The fact that $C = 4.4 \pm 0.6$ is derived from two independent data sources — galactic rotation curves and cosmological expansion rate — constitutes a non-trivial consistency check. The uncertainty on C is propagated from both k and a_G .

3.3 Model Comparison

Table 2: Information-criterion comparison: EFC screening vs. simple interpolation.

Model	Parameters	χ^2_{red}	ΔAIC	ΔBIC
EFC screening ($k, g^\dagger, \sigma_{\text{int}}$)	3	—	0 (ref.)	0 (ref.)
MOND simple (a_0)	1	—	> 0	> 0

The EFC screening form provides a better fit than the MOND simple interpolation function, as measured by both AIC and BIC, despite having two additional parameters. The improvement is driven by the flexibility of the logarithmic screening form at intermediate accelerations where the simple interpolation function is too rigid.

4 Weak Gravitational Lensing: KiDS-1000

4.1 Regime-Activated Lensing Response

At cosmological scales, the EFC screening mechanism manifests as a lensing response $\Sigma(k, z)$ — the cosmological analog of $\mu(g_{\text{bar}})$ at galaxy scales. The modified lensing power spectrum takes the form:

$$P_{\text{EFC}}(k, z) = P_{\Lambda\text{CDM}}(k, z) \times \Sigma^2(k, z), \quad (4)$$

where $\Sigma(k, z)$ activates preferentially at low redshift and on scales where entropy gradients are significant.

4.2 KiDS-1000 Results

Fitting the regime-activated lensing response to the KiDS-1000 cosmic shear two-point correlation functions (1006 sq. deg.; Asgari et al. 2021 [3]), we find:

Table 3: KiDS-1000 lensing fit: Case A (phenomenological).

Quantity	EFC (Case A)	ΛCDM
α_{L2} (lensing coupling)	0.10 ± 0.01	—
$\Delta(-2 \ln \mathcal{L})$ vs. ΛCDM	−50.9	0 (ref.)
S_8	0.685	0.739
Additional parameters	+3	6 (standard)

The improvement $\Delta(-2 \ln \mathcal{L}) = -50.9$ is substantial and localised to low- z bins ($3.2\times$ stronger than high- z), consistent with the EFC prediction of late-time activation.

4.3 S_8 Tension: Updated Landscape

A critical finding is that EFC Case A prefers a lower $S_8 = 0.685$ compared to the ΛCDM value of 0.739, which increases the Planck tension from 2.3σ to 3.6σ . This means Case A is a phenomenological lensing boost, not an S_8 tension solution. A full modified-gravity treatment (Case B, with μ in the growth equations) is required for tension resolution.

Since submission, the Stage-III weak-lensing landscape has shifted upward. Updated values from KiDS Legacy [4] and HSC Y3 [5] are summarised below:

Table 4: Updated Stage-III S_8 landscape (2025–2026).

Probe	S_8	Reference
KiDS Legacy (cosmic shear)	$0.815^{+0.016}_{-0.021}$	Wright et al. (2025)
HSC Y3 (DESI-calibrated $n(z)$)	0.805 ± 0.018	Choppin de Janvry et al. (2025)
Combined WL (KiDS+DES+HSC)	$0.813^{+0.009}_{-0.010}$	Choppin de Janvry et al. (2025)
CMB lensing (combined)	0.828 ± 0.012	Choppin de Janvry et al. (2025)
EFC background coupling ($\beta = 0.08$)	$\sigma_8 \approx 0.805$	This work

The EFC background-coupling prediction ($\sigma_8 \approx 0.805$) is consistent with these updated values without tuning. The S_8 tension motivation should be read in light of this convergence; structural predictions and falsification criteria remain unchanged.

5 Cluster Merger Geometry: The Bullet Cluster

5.1 The Entropy-Gradient Challenge

The Bullet Cluster (1E 0657-56) presents a well-known challenge for any gravity theory that modifies Newtonian dynamics without dark matter: the weak-lensing mass peak is offset from the hot X-ray gas by ~ 150 kpc, coinciding instead with the collisionless galaxy distribution. In MOND-type theories, the gravitational potential should trace the dominant baryonic mass (the gas), predicting the lensing peak at the wrong location.

5.2 EFC Interpretation

In the EFC framework, the effective gravitational potential includes an entropy-gradient contribution:

$$\Phi_{\text{eff}}(\mathbf{r}) = \Phi_N(\mathbf{r}) + \alpha \cdot \nabla S(\mathbf{r}), \quad (5)$$

where Φ_N is the Newtonian potential, α is the EFC coupling constant, and $\nabla S(\mathbf{r})$ is the local entropy gradient.

During a cluster merger, the hot gas undergoes shock heating, generating steep entropy gradients in the inter-cluster medium. The galaxies, being collisionless, retain their pre-merger entropy profile. The EFC prediction is that the lensing signal follows the entropy-weighted potential, not the mass-weighted potential. In the post-merger state:

- The shocked gas has high entropy but steep negative gradients (entropy decreasing away from the shock front), which reduce the effective potential enhancement.
- The galaxy sub-clusters retain moderate entropy with gentle gradients, which maintain the screening-enhanced potential.

This produces a lensing offset in the correct direction — toward the galaxy concentrations — without invoking dark matter. The separation ~ 150 kpc is set by the shock-front geometry and the entropy gradient length scale.

5.3 Epistemic Status

This interpretation is structural and qualitative. A quantitative prediction requires hydrodynamic simulations with the EFC entropy-potential coupling, which constitutes a key next step (Section 7).

6 The Hubble Tension

6.1 Entropy-Modified Friedmann Equation

In EFC, the effective gravitational constant is modulated by the entropy-driven coupling:

$$G_{\text{eff}}(S) = G_0 \exp(a_G \cdot \Delta\Phi(S)), \quad (6)$$

where $a_G \approx 0.094$ is the universal coupling coefficient. From the Friedmann equation $H^2 \propto G_{\text{eff}} \cdot \rho$, taking logarithms and differentiating yields:

$$a_{H_0} = \frac{1}{2} a_G. \quad (7)$$

This factor of $\frac{1}{2}$ is not fitted — it emerges directly from the Friedmann equation structure.

6.2 Derivation of a_G from Observed H_0 Values

From the observed H_0 values (SH0ES: 73.0 km/s/Mpc; Planck: 67.4 km/s/Mpc):

$$\ln(73.0/67.4) = 0.0798, \quad (8)$$

$$a_{H_0} = 0.0798/1.71 = 0.0467, \quad (9)$$

$$a_G = 2 \times a_{H_0} = 0.0935 \approx 0.094. \quad (10)$$

This independently derived $a_G = 0.094$ feeds into the phase-gain parameter $C = k/a_G = 0.415/0.094 = 4.4$, closing the galactic-cosmological loop.

6.3 Closure Conjectures

We propose two testable closure relations [9]:

$$g^\dagger = \frac{c H_0}{e}, \quad (11)$$

$$C = 2e - 1 \approx 4.44. \quad (12)$$

The first relation connects the galactic transition scale to the Hubble rate through Euler's number e , motivated by the Core Lock exponential structure $f(S) = f_0 \exp(-S/\Delta S)$. Using the SPARC-derived $g^\dagger = (2.51 \pm 0.60) \times 10^{-10} \text{ m/s}^2$, this yields $H_0 = 70.2 \pm 16.8 \text{ km/s/Mpc}$ — consistent with all current measurements but not yet precise enough to discriminate between them.

Epistemic status: Conjectured closure. Consistent with data but not derived from first principles. Testable with improved $g^\dagger$ precision.

7 Discussion

7.1 Repositioning in the Stage-III Era

The convergence of Stage-III weak-lensing results shifts the EFC positioning from “alternative model that explains a tension” to “alternative model that matches standard cosmology at first order and predicts second-order structural deviations.” The relevant question is no longer whether S_8 is discrepant, but whether weak-lensing data prefer scale-dependent freedom in the matter-lensing response.

Strengths in the new landscape include: (a) $\beta = 0.08 \rightarrow \sigma_8 = 0.805$ matches converging data without tuning; (b) growth suppression via Hubble friction operates with the correct mechanism and direction; (c) one transition function $T(z)$ governs all regimes without patches; (d) five pre-registered falsification criteria, none triggered.

7.2 Falsification Criteria

The following conditions would falsify the EFC screening model:

1. k outside $[0.30, 0.55]$ from independent SPARC re-analysis.
2. $g^\dagger$ inconsistent with cH_0/e at $>3\sigma$ with improved precision.
3. No lensing response $\Sigma(k, z)$ detected in Stage-IV data (Euclid, LSST).
4. $C = k/a_G$ inconsistent with $2e - 1$ at $>3\sigma$.
5. Bullet Cluster lensing offset explained without entropy-gradient coupling.

7.3 Limitations

- (i) The Bullet Cluster analysis (Section 5) is qualitative; quantitative predictions require hydrodynamic simulations with EFC entropy-potential coupling.
- (ii) The KiDS-1000 fit (Section 4) uses a phenomenological lensing boost (Case A); a self-consistent modified-gravity treatment (Case B) is needed.
- (iii) The closure conjectures (Section 6) are not derived from first principles.
- (iv) A self-consistent EFC background expansion history (not Λ CDM-calibrated) remains to be developed.

7.4 Next Steps

Key discriminating tests for the Stage-IV era include: environment-dependent clustering, scale-dependent growth, lensing response versus dynamical mass in galaxy clusters, and cluster shear profiles. These are domains where EFC predicts deviations from Λ CDM beyond first-order amplitude matching.

8 Conclusion

We have presented a multi-scale empirical validation of the Energy-Flow Cosmology screening model. The principal results are:

1. **SPARC RAR:** The EFC screening form $\ln \mu = k \ln(1 + g^\dagger/g_{\text{bar}})$ fits 174 galaxies (3264 data points) with $k = 0.415 \pm 0.029$, $g^\dagger = (2.51 \pm 0.60) \times 10^{-10} \text{ m/s}^2$, and $\sigma_{\text{int}} \approx 0.14 \text{ dex}$.
2. **Cross-scale consistency:** The phase-gain $C = k/a_G = 4.4 \pm 0.6$ connects galactic and cosmological scales through independently measured parameters.
3. **KiDS-1000:** A regime-activated lensing response improves the fit by $\Delta(-2 \ln \mathcal{L}) = -50.9$ vs. ΛCDM , localised to low- z .
4. **Bullet Cluster:** The EFC entropy-gradient interpretation produces lensing offsets in the correct direction without dark matter (qualitative).
5. **Hubble tension:** The universal coupling $a_G = 0.094$ is independently derived from both galactic (k) and cosmological (H_0) data, with the factor $\frac{1}{2}$ emerging from the Friedmann equation.

The EFC framework provides a consistent, falsifiable description from kpc to Gpc scales. Five explicit falsification criteria are stated, none currently triggered.

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